

Sim-to-Real Transfer in Deep Reinforcement Learning for Bipedal Locomotion

Lingfan Bao, Tianhu Peng, and Chengxu Zhou

Abstract—This chapter addresses the critical challenge of simulation-to-reality (sim-to-real) transfer for deep reinforcement learning (DRL) in bipedal locomotion. After contextualizing the problem within various control architectures, we dissect the “curse of simulation” by analyzing the primary sources of sim-to-real gap: robot dynamics, contact modeling, state estimation, and numerical solvers. Building on this diagnosis, we structure the solutions around two complementary philosophies. The first is to shrink the gap through model-centric strategies that systematically improve the simulator’s physical fidelity. The second is to harden the policy, a complementary approach that uses in-simulation robustness training and post-deployment adaptation to make the policy inherently resilient to model inaccuracies. The chapter concludes by synthesizing these philosophies into a strategic framework, providing a clear roadmap for developing and evaluating robust sim-to-real solutions.

Index Terms—Deep Reinforcement Learning, Sim-to-Real Transfer, Bipedal Locomotion

I. INTRODUCTION

Bipedal robots, machines that walk on two legs, are compelling platforms for operation in human-centric and natural environments. They can climb stairs, step over irregular obstacles, traverse narrow passages, and access spaces that are impractical for wheeled platforms. Their anthropomorphic form factor also enables natural interaction with tools and infrastructure designed for humans, making them suitable for disaster response, healthcare, logistics, and industrial applications.

Bipedal locomotion remains challenging because of its high dimensionality, underactuation, and intermittent contacts. Model-based methods struggle with complex dynamics, whereas deep reinforcement learning (DRL) has achieved impressive simulation results in bipedal locomotion through trial and error. As shown in Fig. 1, DRL achieves more robust performance than model-based control, particularly as task complexity increases. Most controllers adopt either end-to-end policies that map observations to actions or hierarchical policies that decouple high-level (HL) intent from low-level (LL) execution. Both approaches perform well in simulation but transfer unreliably to hardware, a limitation known as the sim-to-real gap.

This transfer difficulty raises three framing questions: *Why do we train in simulation before deployment?* *What precisely causes the sim-to-real gap?* *If this workflow is necessary, how can we improve it?* These questions, together with our control architectures, define the scope of this chapter.

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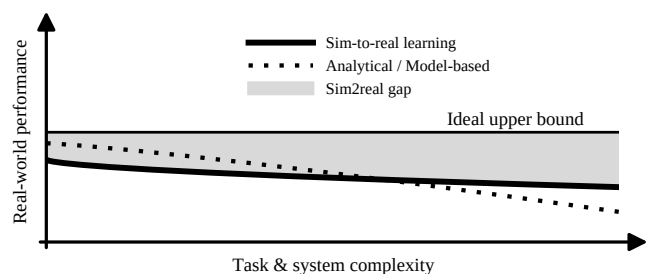


Fig. 1. Comparison of model-based control and sim-to-real learning in terms of real-world performance. As task and system complexity increase, performance declines; however, sim-to-real learning (solid) maintains higher levels than analytical or model-based methods (dotted). The shaded region indicates the sim-to-real gap relative to the ideal upper bound determined by hardware, sensing, and safety constraints.

To begin with the first question, learning locomotion behaviors requires millions of trials and inevitably explores unsafe or costly failure states on hardware. In contrast, simulation enables high-throughput parallel rollouts, precise control of conditions, and reproducibility, making data collection far more efficient and easier to diagnose. Consequently, the practical workflow is to train in a high-fidelity simulator and then transfer the resulting policy to the robot.

Addressing the sim-to-real challenge requires two complementary strategies. The first is to reduce the gap by improving simulator fidelity and calibrating unknown parameters. The second is to strengthen the policy so that it can tolerate residual discrepancies, providing robustness to environmental variations and the ability for online adaptation. In the remainder of this chapter, we organize these strategies into three levers: pre-training alignment, robustness training, and post-deployment adaptation.

This chapter provides a structured overview of sim-to-real transfer for bipedal locomotion with DRL. Section II reviews control architectures, comparing end-to-end policies with hierarchical schemes. Section III-B analyzes the primary sources of the simulation-to-reality (sim-to-real) gap. Section III presents bridging strategies in terms of three levers: pre-training alignment, robustness training, and post-deployment adaptation. Section IV discusses the limitations of current sim-to-real methods, distills lessons from theory to practice, and offers a future outlook. Section V concludes with practical guidelines and open research problems.

面向双足运动的深度强化学习中的仿真到现实迁移

Lingfan Bao, Tianhu Peng, 和 Chengxu Zhou

摘要——本章探讨了在双足运动中，深度强化学习（DRL）从仿真到现实（sim-to-real）迁移所面临的关键挑战。在将问题置于各种控制架构的背景下进行阐述后，我们通过分析仿真到现实差距的主要来源：机器人动力学、接触建模、状态估计和数值求解器，来剖析“仿真诅咒”。基于这一诊断，我们围绕两种互补的理念构建解决方案。第一种是通过以模型为中心的策略来缩小差距，这些策略系统地提高了仿真器的物理保真度。第二种是强化策略，这是一种互补的方法，通过仿真内鲁棒性训练和部署后自适应，使策略本身对模型不准确具有内在的鲁棒性。本章最后将这些理念综合成一个战略框架，为开发和评估鲁棒的仿真到现实解决方案提供了清晰的路线图。

索引术语——深度强化学习，仿真到现实迁移，双足运动

1. 引言

双足机器人，即用两条腿行走的机器，是在以人为中心和自然环境中运行的极具吸引力的平台。它们能够爬楼梯、跨越不规则障碍物、穿越狭窄通道，并进入轮式平台难以企及的空间。其拟人形态还能与为人类设计的工具和基础设施进行自然交互，使其适用于灾难响应、医疗保健、物流和工业应用。

双足运动因其高维性、欠驱动和间歇性接触而仍然具有挑战性。基于模型的方法难以处理复杂的动力学，而深度强化学习通过试错在双足运动仿真中取得了令人瞩目的成果。如图1所示，深度强化学习比基于模型的控制实现了更鲁棒的性能，尤其是在任务复杂度增加时。大多数控制器采用将观测映射到动作的端到端策略，或将高层意图与低层执行解耦的分层策略。这两种方法在仿真中表现良好，但迁移到硬件时可靠性不足，这一局限性被称为

仿真到现实差距。

这种迁移困难引出了三个框架性问题：为什么我们在部署前要在仿真中训练？究竟是什么导致了仿真到现实差距？如果这一工作流程是必要的，我们该如何改进？这些问题，连同我们的控制架构，共同界定了本章的范畴。

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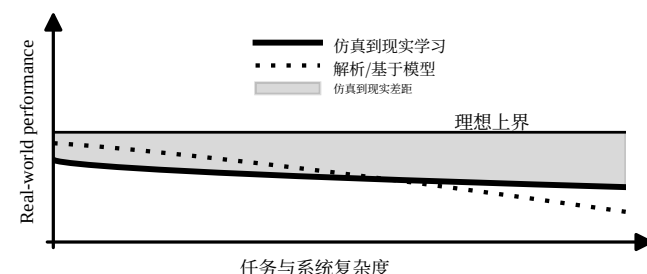


图1. 基于模型的控制与仿真到现实学习在实际性能上的对比。随着任务和系统复杂度的增加，性能会下降；然而，仿真到现实学习（实线）相比解析法或基于模型的方法（虚线）能维持更高的性能水平。阴影区域表示相对于由硬件、感知和安全约束所决定的理想上限的仿真到现实差距。

首先回答第一个问题，学习运动行为需要数百万次试验，并且不可避免地会在硬件上探索不安全或代价高昂的失败状态。相比之下，仿真能够实现高吞吐量的并行运行、精确控制条件以及可重复性，使得数据收集更加高效且易于诊断。因此，实际的工作流程是在高保真仿真器中训练，然后将生成的策略迁移到机器人上。

应对仿真到现实挑战需要两种互补策略。第一种是通过提升仿真器保真度并校准未知参数来缩小差距。第二种是强化策略，使其能够容忍残余差异，从而具备对环境变化的鲁棒性以及在线自适应能力。在本章剩余部分，我们将这些策略组织为三个杠杆：预训练对齐、鲁棒性训练和部署后自适应。

本章对基于深度强化学习的双足运动仿真到现实迁移进行了结构化概述。第二部分回顾了控制架构，比较了端到端策略与分层方案。第三部分-B分析了仿真到现实差距的主要来源。第三部分从三个杠杆角度介绍了桥接策略：预训练对齐、鲁棒性训练和部署后适应。第四部分讨论了当前仿真到现实方法的局限性，从理论到实践提炼经验教训，并提供了未来展望。第五部分以实用指南和开放研究问题作为总结。

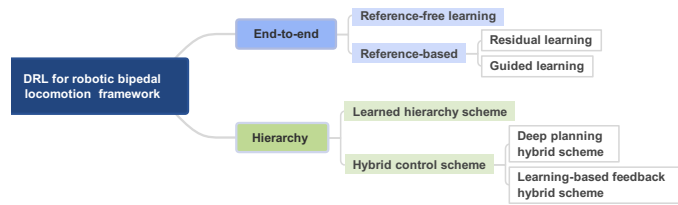


Fig. 2. **Classification of DRL-based control schemes.**

II. CONTROL ARCHITECTURES

Sim-to-real DRL for bipedal locomotion is commonly organized into two control architectures: end-to-end and hierarchical. End-to-end policies map observations directly to low-level actions (for example, torques or impedance targets), allowing the robot to discover viable behaviors without manually specified intermediate structures. Hierarchical architectures, in contrast, decompose decision making into multiple layers. The two categories are summarized in Fig. 2. The choice between them represents a trade-off between simplicity and expressive capacity on one side, and interpretability, explicit constraint handling, and composability on the other, with significant implications for sim-to-real transfer.

The remainder of this section contrasts these approaches and their policy representations, highlights assumptions and limitations, and provides practical guidance for sim-to-real deployment.

A. End-to-End Framework

Formally, an end-to-end DRL policy $\pi : \mathcal{X}! \rightarrow \mathcal{U}$ maps sensory inputs $\mathcal{X}$, such as images, lidar, or proprioceptive feedback [1], together with user commands [2] or predefined references [3], to joint-level actions $\mathcal{U}$. Here, $\mathcal{X}$ denotes the observation space, $\mathcal{U}$ the action space, and $\pi(\cdot)$ the policy. In practice, lightweight state estimation and safety filters are typically retained.

End-to-end methods can be categorized by the amount of prior structure they employ. Reference-based policies condition on motion templates or commanded trajectories, which improves sample efficiency and targeting but introduces biases and limits feasibility. Residual or guided variants can help mitigate these drawbacks. Reference-free policies, by contrast, learn purely from task rewards, providing greater expressiveness but relying heavily on reward shaping, exploration, and robustness for successful transfer. Figure 3 summarizes three representative training strategies.

a) Reference-Based Learning: Reference-based learning leverages prior knowledge generated offline through methods such as trajectory optimization (TO) or motion capture systems. This predefined reference typically contains data related to the robot’s joint movements or preplanned trajectories, providing a foundation for the policy to develop locomotion skills by following established motion patterns. In general, this approach can be divided into two primary methods: (i) residual learning and (ii) guided learning.

The residual learning framework employs a policy that adjusts motor commands by applying action offsets relative to

the current reference joint positions. This enables the bipedal robot to achieve dynamic locomotion through error compensation. The state space includes proprioceptive information such as trunk position, orientation, velocity, angular velocity, joint angles, and joint velocities, which together provide the necessary sensory data for real-time adjustments. Actions are defined by offsets δa , representing deviations from the desired joint positions $\hat{a}$, and the final motor commands are given by $a = \hat{a} + \delta a$. The reward function encourages the policy to optimize locomotion performance by considering (a) how closely the robot’s active joint angles match the reference angles, (b) how effectively the robot responds to user commands, and (c) additional terms that enhance stability and smoothness of motion. This formulation allows the robot to adapt to dynamic conditions while maintaining balance and control.

Guided learning, in contrast, trains the policy to directly output joint-level commands as actions a without adding a residual term. The state space is the same as that used in residual learning. In this approach, the reward structure focuses on accurately imitating predefined reference trajectories, ensuring close alignment between the policy outputs and the reference motion.

b) Reference-Free Learning: In reference-free learning, the policy is trained using a carefully designed reward function rather than predefined trajectories. This approach allows the policy to explore a wider range of gait patterns and adapt to unpredictable terrains, thereby enhancing flexibility and innovation in the learning process. The action and observation spaces are similar to those in the guided-learning framework; however, the reward structure differs substantially. Instead of emphasizing imitation of reference motions, the reward focuses on developing efficient gait patterns that capture the distinctive characteristics of bipedal locomotion [4].

The concept of reference-free learning was initially explored in simulated physics environments using simplified bipedal models. A pioneering framework demonstrated that symmetric gaits could be learned entirely from scratch without motion capture data [5]. This framework introduced a specialized term in the loss function and employed curriculum learning to gradually shape the gait. Another significant contribution involved a learning method that enabled a robot to traverse stepping stones through curriculum-based training on the Cassie platform, although this method has not yet been validated outside of simulation [6].

This learning paradigm promotes the discovery of novel strategies and behaviors that may not emerge through imitation alone. However, the absence of reference guidance can make training costly, time-consuming, and occasionally infeasible for complex tasks. Furthermore, the success of reference-free learning depends heavily on the design of the reward function, which poses particular challenges for dynamic behaviors such as jumping.

B. Hierarchical Framework

Unlike end-to-end policies that directly map sensory inputs to motor commands, hierarchical control decomposes locomotion into multiple decision stages. A canonical architecture

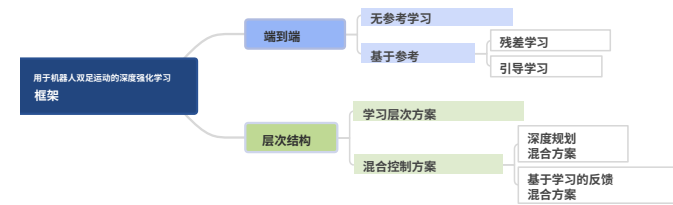


图 2. 基于深度强化学习的控制方案分类

II. 控制架构

用于双足运动的仿真到现实深度强化学习通常组织为两种控制架构：端到端和分层。端到端策略将观测直接映射到低级动作（例如，力矩或阻抗目标），使机器人能够发现可行的行为，而无需手动指定中间结构。相比之下，分层架构将决策分解为多个层次。图2总结了这两类架构。两者之间的选择代表了简单性与表达能力一侧，与可解释性、显式约束处理和可组合性另一侧之间的权衡，这对仿真到现实迁移具有重要影响。

本节其余部分将对对比这些方法及其策略表示，强调各自的假设与局限性，并为仿真到现实的部署提供实用指导。

A. 端到端框架

形式上，一个端到端深度强化学习策略 $\pi : \mathcal{X}! \rightarrow \mathcal{U}$ 将感官输入 $\mathcal{X}$ （如图像、激光雷达或本体感觉反馈 [1]）与用户指令 [2] 或预定义参考 [3]，一起映射到关节级动作 $\mathcal{U}$ 。这里， $\mathcal{X}$ 表示观测空间， $\mathcal{U}$ 表示动作空间， $\pi(\cdot)$ 表示策略。在实践中，轻量级状态估计和安全过滤器通常会被保留。

端到端方法可根据其采用的先验结构量进行分类。基于参考的策略以运动模板或指令轨迹为条件，这提高了样本效率和目标针对性，但引入了偏差并限制了可行性。残差或引导变体有助于缓解这些缺点。相比之下，无参考策略纯粹从任务奖励中学习，提供了更强的表达能力，但严重依赖奖励塑形、探索和鲁棒性以实现成功迁移。图3总结了三种代表性的训练策略。

*a) 基于参考的学习：*基于参考的学习利用了通过轨迹优化（TO）或动作捕捉系统等离线方法生成的先验知识。这种预定义参考通常包含与机器人关节运动或预规划轨迹相关的数据，为策略通过遵循既定运动模式来发展运动技能提供了基础。通常，这种方法可分为两种主要方式：(i) 残差学习和 (ii) 引导学习。

残差学习框架采用一种策略，该策略通过施加相对于当前参考关节位置的动作偏移来调整电机指令。

这使得双足机器人能够通过误差补偿实现动态运动。状态空间包括本体感觉信息，如躯干位置、朝向、速度、角速度、关节角度和关节速度，这些信息共同为实时调整提供了必要的感知数据。动作由偏移量 δa 定义，表示与期望关节位置 $\hat{a}$ 的偏差，最终的电机指令由 $a = \hat{a} + \delta a$ 给出。奖励函数通过考虑以下因素来鼓励策略优化运动性能：(a) 机器人的主动关节角度与参考角度的匹配程度，(b) 机器人响应用户指令的有效性，以及 (c) 增强稳定性和运动平滑性的附加项。这种公式化方法使机器人能够在保持平衡和控制的同时适应动态条件。

相比之下，引导学习训练策略直接输出关节级命令作为动作 a ，无需添加残差项。其状态空间与残差学习相同。在这种方法中，奖励结构侧重于精确模仿预定义的参考轨迹，确保策略输出与参考运动紧密对齐。

*b) 无参考学习：*在无参考学习中，策略通过精心设计的奖励函数而非预定义轨迹进行训练。这种方法允许策略探索更广泛的步态模式并适应不可预测的地形，从而增强学习过程的灵活性和创新性。动作空间和观测空间与引导学习框架相似，但奖励结构有显著差异。奖励不再强调模仿参考运动，而是侧重于开发能够捕捉双足运动独特特征的高效步态模式 [4]。

无参考学习的概念最初在模拟物理环境中使用简化双足模型进行探索。一个开创性框架证明，无需动作捕捉数据即可从零开始学习对称步态 [5]。该框架在损失函数中引入了一个专门项，并采用课程学习逐步塑造步态。另一项重要贡献涉及一种学习方法，使机器人能够通过基于课程训练在Cassie平台上穿越踏脚石，尽管该方法尚未在仿真之外得到验证 [6]。

这种学习范式促进了新策略和行为的发现，而这些可能无法仅通过模仿实现。然而，缺乏参考引导会使训练成本高昂、耗时，并且对于复杂任务有时甚至不可行。此外，无参考学习的成功在很大程度上取决于奖励函数的设计，这对于跳跃等动态行为尤其具有挑战性。

B. 分层框架

与直接将感官输入映射到电机指令的端到端策略不同，分层控制将运动分解为多个决策阶段。一种典型架构

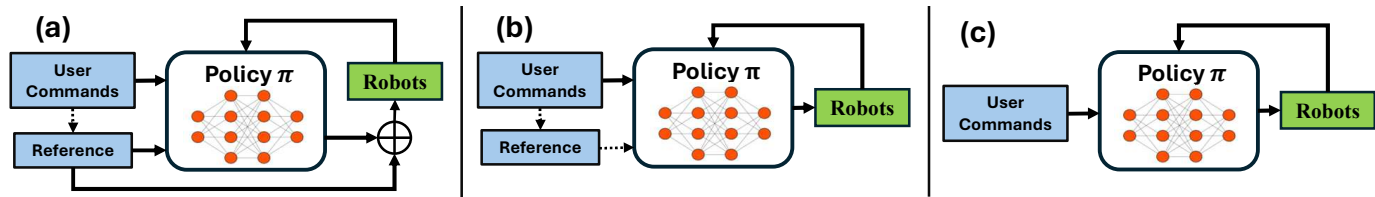


Fig. 3. End-to-end locomotion policy variants. (a) Residual: the policy adds a bounded correction to a nominal controller. (b) Guided: the policy is conditioned on a motion or controller reference. (c) Reference-free: the policy learns solely from task rewards. Feedback arrows indicate onboard estimates returned to the policy.

includes two modules: an HL planner and an LL controller. Either module can be model-based or learned, allowing task-specific customization. We categorize variants into three types, summarized in Fig. 4. The HL planner issues abstract goals that the LL controller translates into executable trajectories and motor commands, while sensor feedback flows upward for online correction. Because the layers operate at different frequencies, communication and synchronization between them are central design considerations. We begin with deep-planning hybrids, in which a learned planner generates task-space setpoints and a model-based controller executes them at the joint level.

a) Deep-Planning Hybrid Scheme: Several studies have demonstrated the integration of a learned HL planner with a model-based controller to achieve task-space objectives. One representative framework optimizes task-space performance without explicitly handling joint-level balance dynamics [7]. This system combines a residual-learning planner with an inverse dynamics controller, enabling precise translation of task-space commands into joint-level actions and improving velocity tracking, foot placement, and height control. Further advancements include hybrid frameworks that merge HZD-based residual deep planning with model-based regulators to correct trajectory errors, demonstrating robustness, training efficiency, and accurate velocity tracking [8]. These approaches have successfully transferred from simulation to physical platforms such as Cassie.

b) Feedback-DRL Control Hybrid Scheme: Unlike the end-to-end formulation in Section II-A, this framework places a DRL policy at the LL while an HL planner provides terrain-aware footstep or center-of-mass (CoM) references and safety margins. Gait-library architectures follow this pattern [9], although static references restrict adaptability to varying terrain. A more flexible design combines a conventional footstep planner with an LL DRL controller that tracks planned steps and headings [10], initially demonstrated in simulation. Later work using a similar HL–LL division achieved successful sim-to-real transfer, enabling omnidirectional walking, obstacle avoidance, and stair negotiation [11]. Injecting HL feedback, such as CoM or end-effector states, into the LL policy further enhances robustness.

In this scheme, robots are typically equipped with model-based feedback controllers or interpretable methods that handle basic locomotion skills such as walking. The addition of a learned HL layer focuses on strategic task-space goals, thereby extending locomotion capabilities and enabling advanced navigation behaviors.

c) Learned Hierarchical Scheme: The learned hierarchy framework integrates a learned HL planner with a learned LL controller. Training generally begins with refining the LL policy to ensure balance and fundamental locomotion, followed by developing an HL policy that directs the robot toward specific targets. This layered process supports structured autonomy and facilitates scalable training.

Early hierarchical systems were validated in simulation [12]. In these systems, the LL controller tracked motion references derived from human demonstrations or TO, while the HL planner issued long-horizon navigation goals, enabling tasks such as soccer dribbling. Subsequent work incorporated imitation learning to reproduce agile, human-like skills in physics simulation [13].

In these learned hierarchies, the HL layer governs overall strategy and goals and can be further refined through continued training. Recent studies report successful sim-to-real implementations on humanoid robots, including human-motion-conditioned whole-body control and outdoor locomotion [14], [15]. However, training multiple layers remains computationally expensive and simulation-centric, which can cause overfitting and poor generalization [16]. Reliable deployment requires clean interfaces and robust feedback between layers.

While hierarchical control improves modularity and interpretability, it also introduces distinct challenges for sim-to-real transfer. The HL planner relies on assumptions about dynamics, terrain, and sensing that may not hold in reality. The LL controller must cope with contact, friction, latency, and actuator limits that differ between simulation and hardware. Moreover, the interface between levels imposes bandwidth and timing constraints that are easily maintained in simulation but fragile on real systems. These effects can compound, as small modeling errors in the planner propagate to aggressive setpoints that the controller executes with mismatched gains or delays.

III. CROSSING SIMULATION AND REALITY

Building on the control schemes introduced in Section II, we now address how policies trained in simulation can be effectively transferred to hardware. Most contemporary bipedal locomotion frameworks follow this train-in-simulation, deploy-on-hardware paradigm. Although transfer requires considerable effort, it remains the most practical and scalable approach for developing real-world controllers.

This section revisits the questions posed in Section I regarding sim-to-real transfer. We begin by explaining why

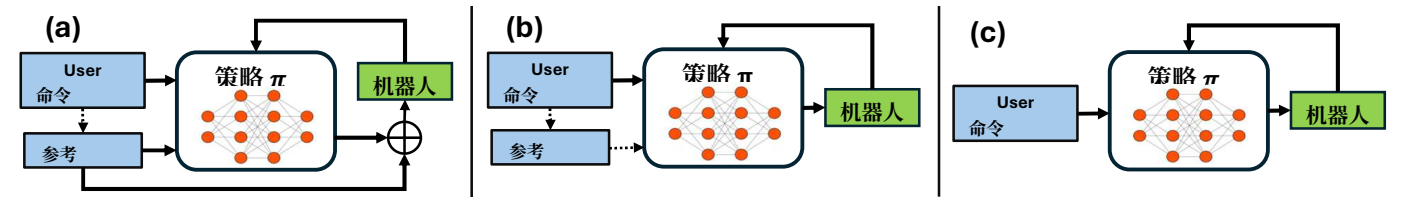


图 3. 端到端运动策略变体。(a) 残差：策略向标称控制器添加有界修正。(b) 引导式：策略以运动或控制器参考为条件。(c) 无参考：策略仅从任务奖励中学习。反馈箭头表示返回给策略的机载估计值。

包含两个模块：一个高层规划器和一个底层控制器。每个模块可以是基于模型的或学习型的，从而允许针对特定任务进行定制。我们将其变体归纳为三种类型，如图4所示。高层规划器发出抽象目标，底层控制器将其转化为可执行轨迹和电机指令，同时传感器反馈向上流动以进行在线校正。由于各层以不同频率运行，它们之间的通信与同步是核心设计考量。我们从深度规划混合体开始，其中学习型规划器生成任务空间设定点，而基于模型的控制器在关节层面执行这些设定点。

*a) 深度规划混合方案：*多项研究展示了将学习型高层规划器与基于模型的控制器相结合，以实现任务空间目标。其中一个代表性框架优化了任务空间性能，而无需显式处理关节级平衡动力学 [7]。该系统将残差学习规划器与逆动力学控制器相结合，能够将任务空间命令精确转化为关节级动作，并改善速度跟踪、足部放置和高度控制。进一步的进展包括融合了基于HZD的残差深度规划与基于模型的调节器的混合框架，以纠正轨迹误差，展现出鲁棒性、训练效率和准确的速度跟踪能力 [8]。这些方法已成功从仿真迁移到诸如Cassie等物理平台上。

*b) 反馈深度强化学习控制混合方案：*与第二节A部分中的端到端公式不同，该框架将深度强化学习策略置于底层，而高层规划器则提供地形感知的步态或质心参考以及安全裕度。步态库架构遵循此模式 [9]，尽管静态参考限制了其对变化地形的适应性。一种更灵活的设计将传统步态规划器与底层深度强化学习控制器相结合，用于跟踪规划的步态和航向 [10]，最初在仿真中得到验证。后续工作采用类似的高层-底层划分，成功实现了从仿真到现实的迁移，实现了全向行走、避障和楼梯攀爬 [11]。将高层反馈（例如质心或末端执行器状态）注入底层策略，进一步增强了鲁棒性。

在该方案中，机器人通常配备基于模型的反馈控制器或可解释方法，用于处理行走等基本运动技能。学习型高层的加入专注于战略性任务空间目标，从而扩展运动能力并实现高级导航行为。

*c) 学习型分层方案：*学习型分层框架将学习型高层规划器与学习型底层控制器相结合。训练通常从优化底层策略以确保平衡和基本运动开始，随后开发引导机器人朝向特定目标的高层策略。这一分层过程支持结构化自主性，并促进可扩展训练。

早期的分层系统在仿真中得到了验证 [12]。在这些系统中，底层控制器跟踪源自人类演示或任务规划的参考运动，而高层规划器则发布长时域导航目标，从而实现了足球运球等任务。后续工作引入了模仿学习，以在物理仿真中复现敏捷、类人的技能 [13]。

在这些学习型分层结构中，高层负责整体策略和目标，并可通过持续训练进一步优化。近期研究报告了在人形机器人上成功的仿真到现实实现，包括基于人体运动条件的全身控制和户外运动 [14][15]。然而，训练多个层级的计算成本仍然很高，且以仿真为中心，这可能导致过拟合和泛化能力差 [16]。可靠的部署需要各层之间具备清晰接口和鲁棒反馈。

虽然分层控制提高了模块化和可解释性，但它也为仿真到现实的迁移带来了独特的挑战。高层规划器依赖于对动力学、地形和感知的假设，这些假设在现实中可能不成立。底层控制器必须应对仿真与硬件之间不同的接触、摩擦、延迟和执行器限制。此外，层级之间的接口施加了带宽和时序约束，这些约束在仿真中易于维持，但在真实系统中却很脆弱。这些影响可能会叠加，因为规划器中的微小建模误差会传播到激进的设定点，而控制器则使用失配增益或延迟来执行这些设定点。

III. 跨越仿真与现实

基于第二部分介绍的控制方案，我们现在探讨如何将仿真中训练的策略有效迁移至硬件。大多数现代双足运动框架都遵循这种仿真训练、硬件部署范式。尽管迁移需要付出相当大的努力，但它仍然是开发真实世界控制器最实用且可扩展的方法。

本节重新审视了第一节中关于仿真到现实迁移所提出的问题。我们首先解释为何

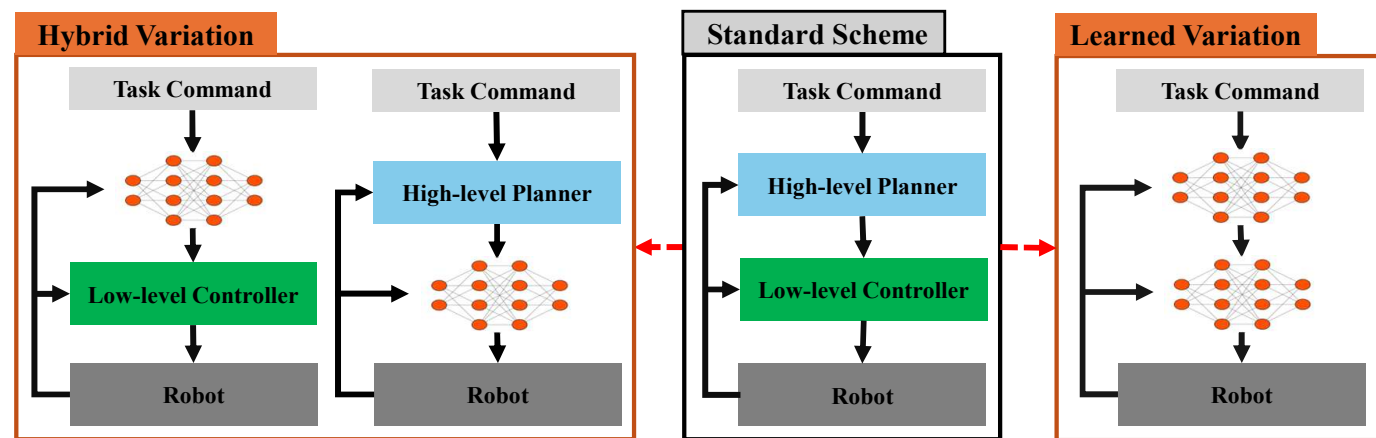


Fig. 4. Hierarchical control architectures for a bipedal robot. The central panel (“Standard Scheme”) presents the canonical hierarchy: a task command is processed by an HL planner, whose output is executed by an LL controller to actuate the robot. The left panel (“Hybrid Variation”) illustrates configurations in which one layer is learned while the other remains model-based (either a learned HL planner with a classical LL controller or the reverse). The right panel (“Learned Variation”) implements both HL planning and LL control as learned policies arranged in a two-layer hierarchy.

training in simulation is generally preferred. We then identify the principal sources of the sim-to-real gap and discuss their implications for improving simulator fidelity. Finally, we outline a pragmatic strategy that balances accuracy and robustness through three complementary components: (i) pre-training alignment, which improves simulator fidelity and encodes nonprivileged information into the policy; (ii) domain randomization (DR) with curriculum learning, which spans plausible uncertainties during training; and (iii) online adaptation, which enables the robot to handle environmental changes at deployment. Figure 5 summarizes this roadmap.

A. Why Sim-to-Real Training?

Training locomotion policies directly on physical bipedal robots is prohibitively expensive and unsafe [17]. DRL typically requires tens of millions of interactions to produce stable gaits, which would correspond to months of continuous operation if performed entirely on hardware. Such prolonged trial-and-error accelerates mechanical wear, increases the likelihood of catastrophic failures, and demands continuous human supervision for resets and safety monitoring. On-robot learning is further constrained by partial observability, the difficulty of designing informative reward functions, and the rarity of safety-critical disturbances that are essential for robustness.

Simulation offers a practical alternative. Modern physics engines support massively parallel and faster-than-real-time rollouts, enabling policies to accumulate experience orders of magnitude faster than in the real world. Simulators also allow safe exploration of failure cases, controlled curricula that progressively increase task difficulty, full access to ground-truth states for dense rewards and diagnostics, and reproducible experiments across research groups.

Consequently, nearly all modern locomotion research adopts a sim-to-real workflow: policies are first trained in simulation, then adapted, calibrated, or otherwise transferred to hardware. This paradigm balances data efficiency with safety while motivating the transfer methodologies discussed throughout this chapter.

B. The Curse of Simulation

To understand why transfer remains challenging, we must examine why even high-fidelity simulators deviate from reality. Despite detailed models of the robot, terrain, and environment, residual discrepancies persist. These mismatches arise primarily from modeling errors in robot dynamics and actuation, interactions with the environment, terrain compliance, and sensing. Additional deviations originate from numerical artifacts introduced by discretization, integration schemes, and contact solvers. The remainder of this subsection details each source and its practical implications.

a) Robot Dynamics Modeling and Actuation: The robot’s dynamic model is a primary contributor to the sim-to-real gap. The rigid-body dynamics, expressed in the Lagrangian form in Equation 1, highlight the major components where modeling errors can arise:

$$\underbrace{M(q)\ddot{q}}_{\text{Inertia term}} + \underbrace{h(q, \dot{q})}_{\text{Coriolis, gravity}} = \underbrace{S^\top \tau}_{\text{Actuation torques}} + \underbrace{J_c(q)^\top \lambda}_{\text{Contact forces}} + \underbrace{\tau_{\text{ext}}}_{\text{External disturbance}} \quad (1)$$

where $M(q)$ is the mass matrix, $h(q, \dot{q})$ aggregates Coriolis, centrifugal, and gravity effects, $S^\top \tau$ represents actuation torques, and $J_c^\top \lambda$ represents contact wrenches. High-fidelity simulation begins with consistent coordinate frames and accurate inertial parameters for each link, including the mass m , center of mass r_{com} , and inertia tensor I_{com} expressed about the CoM in the link frame.

For bipedal robots, whose long kinematic chains and floating-base dynamics amplify small discrepancies, inaccuracies in these parameters are particularly consequential. Misestimated inertial properties distort the torques required for balance and gait generation, leading to instability and reduced energy efficiency. Therefore, precise identification of rigid-body parameters is often the most effective first step in narrowing the sim-to-real gap.

Beyond rigid-body modeling, the actuation system represents another major source of mismatch. Simulators frequently

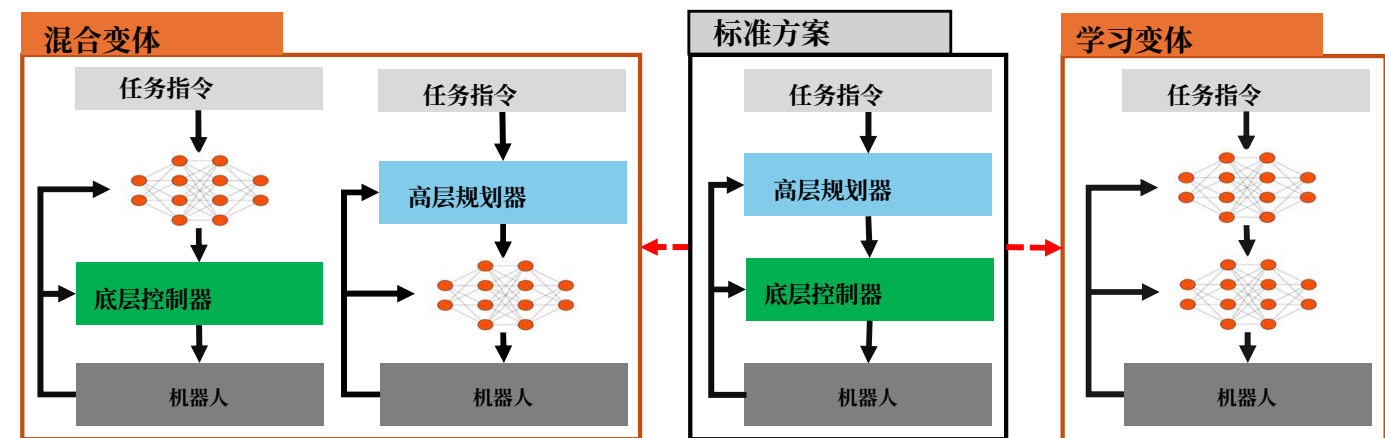


图4. 双足机器人的分层控制架构。中央面板（“标准方案”）展示了经典层次结构：任务指令由高层规划器处理，其输出由底层控制器执行以驱动机器人。左侧面板（“混合变体”）展示了其中一层为学习型而另一层保持基于模型的配置（即学习型高层规划器搭配经典底层控制器，或反之）。右侧面板（“学习变体”）则将高层规划与底层控制均实现为学习策略，并排列成两层层次结构。

在仿真中进行训练通常是首选。接着，我们识别出仿真到现实差距的主要来源，并讨论其对提升仿真器保真度的启示。最后，我们概述了一种通过三个互补组件平衡精度与鲁棒性的实用策略：(i) 预训练对齐，用于提升仿真器保真度并将非特权信息编码到策略中；(ii) 结合课程学习的域随机化，在训练过程中覆盖合理的不确定性；(iii) 在线自适应，使机器人能够在部署时应对环境变化。图5总结了这一路线图。

A. 为何采用仿真到现实训练？

直接在物理双足机器人上训练运动策略成本过高且不安全 [17]。深度强化学习通常需要数千万次交互才能产生稳定步态，若完全在硬件上执行，这相当于连续运行数月。如此长时间的试错会加速机械磨损，增加灾难性故障的可能性，并且需要持续的人工监督以进行重置和安全监控。在机器人本体上进行学习还受到部分可观测性、设计有效奖励函数的难度，以及对于鲁棒性至关重要的安全关键扰动稀缺性的进一步限制。

仿真提供了一种实用的替代方案。现代物理引擎支持大规模并行且超实时的推演，使策略能够以比现实世界快数个数量级的速度积累经验。仿真器还允许安全地探索失败案例、通过逐步增加任务难度的受控课程进行训练、完全访问真实状态以获取密集奖励和诊断信息，并在不同研究团队之间实现可重复的实验。

因此，几乎所有现代运动研究都采用仿真到现实 workflow：策略首先在仿真中训练，然后进行适配、校准或以其他方式迁移到硬件上。这种范式在兼顾数据效率与安全性的同时，也推动了本章所讨论的各种迁移方法。

B. 仿真诅咒

要理解迁移为何仍具挑战性，我们必须探究即使是高保真仿真器为何也会偏离现实。尽管对机器人、地形和环境进行了详细建模，残余差异依然存在。这些失配主要源于机器人动力学与驱动中的建模误差、与环境的交互、地形顺应性以及感知。额外的偏差来自离散化、积分方案和接触求解器引入的数值伪影。本小节的其余部分将详细阐述每个来源及其实际影响。

a) 机器人动力学建模与驱动： 机器人的动力学模型是仿真到现实差距的主要来源。以拉格朗日形式表达于公式1中的刚体动力学，突出了可能产生建模误差的主要组成部分：

$$\underbrace{M(q)\ddot{q}}_{\text{惯性项}} + \underbrace{h(q, \dot{q})}_{\text{科里奥利力, 重力}} = \underbrace{S^\top \tau}_{\text{驱动力矩}} + \underbrace{J_c(q)^\top \lambda}_{\text{接触力}} + \underbrace{\tau_{\text{ext}}}_{\text{外部扰动}} \quad (1)$$

其中 $M(q)$ 是质量矩阵， $h(q, \dot{q})$ 聚合了科里奥利力、离心力和重力效应， $S^\top \tau$ 表示驱动力矩， $J_c^\top \lambda$ 表示接触力旋量。高保真仿真始于一致的坐标系和每个连杆的精确惯性参数，包括质量 m 、质心 r_{com} 以及在连杆坐标系中关于质心表达的惯性张量 I_{com} 。

对于双足机器人而言，其较长的运动链和浮动基座动力学放大了微小差异，因此这些参数的不准确性尤为关键。惯性参数估计错误会扭曲平衡与步态生成所需的力矩，导致不稳定性并降低能效。因此，精确辨识刚体参数往往是缩小仿真到现实差距最有效的第一步。

除了刚体建模之外，驱动系统是另一个主要的失配来源。模拟器通常

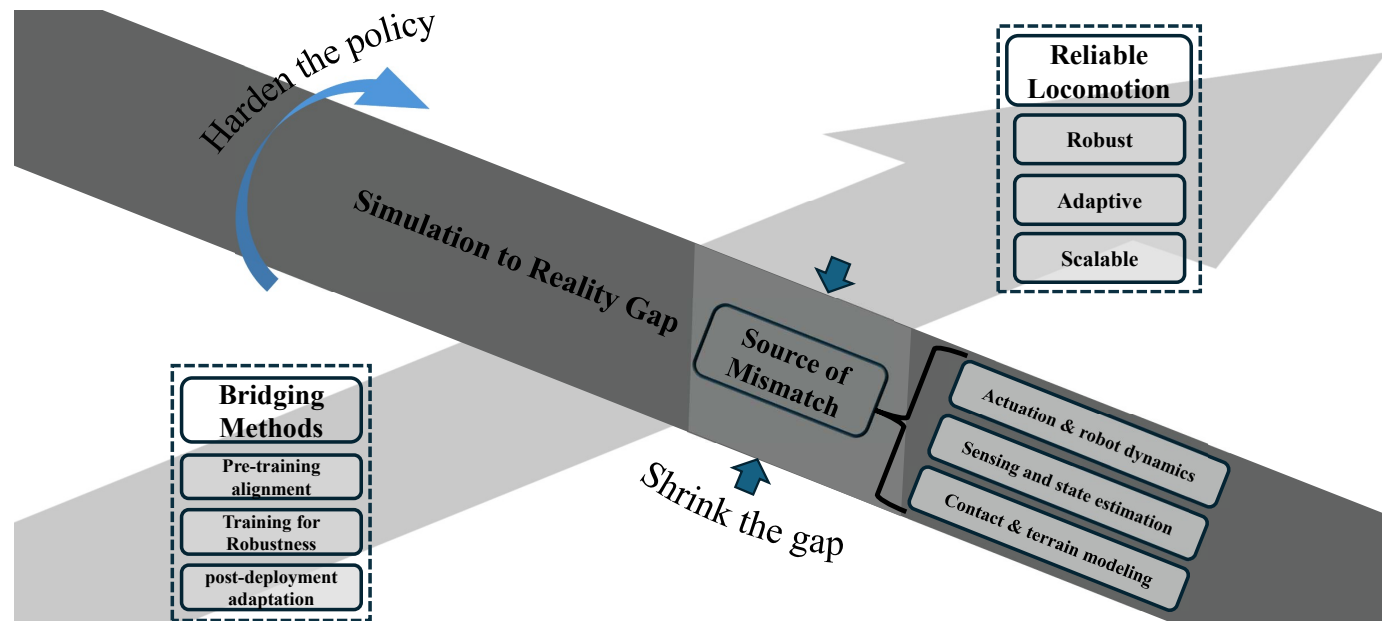


Fig. 5. High-level roadmap for sim-to-real bipedal locomotion. Two levers help reduce the transfer gap: (1) shrinking the gap by improving simulator fidelity and system identification, and (2) hardening the policy through domain randomization (DR) and curriculum learning. Residual discrepancies are managed through online adaptation during deployment. Representative sources of mismatch include actuation and robot dynamics, sensing and state estimation, and contact and terrain modeling. The ultimate goal is to achieve robust, adaptive, and scalable bipedal locomotion.

idealize actuators as perfect torque sources, neglecting real-world nonlinearities that are critical to locomotion. In practice, the actual torque delivered depends on several nonideal effects: (i) Hardware limits, including saturation and slew-rate constraints caused by finite voltage, current, and thermal boundaries [18]; and (ii) Electromechanical dynamics, including finite bandwidth and command delays that effectively filter the input signal.

These effects alter both the timing and magnitude of contact forces and are a primary reason that policies appearing stable in simulation often fail on hardware. The underlying motor dynamics are governed by Equation 2:

$$\dot{i} = \frac{1}{L}(v - Ri - k_e \omega_m), \tau_m = k_t i, P = v i, \quad (2)$$

where the applied voltage v and back electromotive force (EMF) $k_e \omega_m$ determine the current i and resulting motor torque τ_m . As Equation 2 illustrates, these dynamics act as a low-pass filter on torque commands, introducing latency and limiting peak responsiveness. Neglecting this filtering effect during simulation often encourages policies that are overly aggressive or poorly timed, which become unstable when deployed on physical robots. Accurate transfer, therefore, requires explicit modeling of actuator limits and response dynamics calibrated from empirical measurements.

b) Contact and Terrain Modeling: Beyond robot dynamics, the next major contributor to the sim-to-real gap is environment interaction. For bipedal locomotion, policy stability is highly sensitive to the modeled contact physics. Even small inaccuracies in contact stiffness, friction, or terrain geometry can alter ground-reaction forces (GRFs) and gait timing, destabilizing otherwise functional policies [19], [20]. Furthermore, real-world surfaces exhibit nonstationary

variations in friction and compliance due to wear, moisture, or surface texture—factors that most static simulation models neglect.

Simulators typically employ one of two paradigms: computationally expensive rigid-body models that capture sharp stick-slip transitions, or more stable but less accurate compliant models that approximate contact with spring-damper systems. Both approaches introduce artifacts. Poorly tuned compliant models can exaggerate impact forces or artificially damp energy, while simplified friction models can bias balance-critical Center of Pressure (CoP) estimates. Because realistic deformable terrain simulation is computationally costly, training is often restricted to rigid, planar surfaces, resulting in policies that fail on real-world compliant terrain such as carpet or grass.

c) Sensing and State Estimation: In simulation, locomotion policies are typically trained with privileged, noise-free state information, including true joint angles, velocities, and base poses. On hardware, however, these quantities are available only as estimates obtained from encoders, IMUs, and contact sensors through filtering and sensor-fusion pipelines. Such pipelines introduce quantization, bias drift, colored noise, latency, timestamp jitter, and occasional dropouts, rendering the system effectively partially observable. The resulting estimation errors and delays perturb gait phase timing and feedback gains, causing policies trained on perfect state information to become brittle during deployment.

To mitigate these discrepancies, the training setup should mimic the deployed observation interface and timing as closely as possible. Estimator-like inputs should replace ground-truth states, and the statistics of sensing and filtering (noise, bias, latency, dropouts) should be emulated within physically plausible ranges. Policy architectures should integrate information

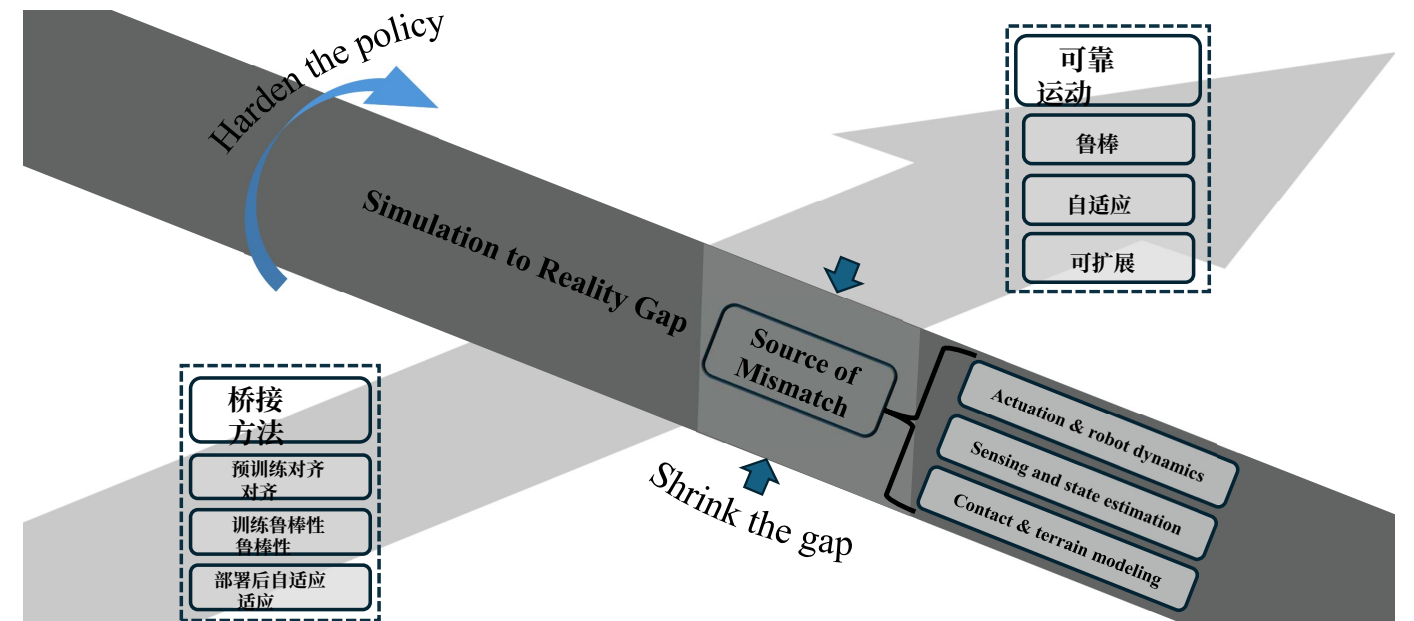


图5. 仿真到现实双足运动的高层路线图。两个杠杆有助于缩小迁移差距：(1) 通过提升仿真器保真度和系统辨识来缩小差距，以及(2) 通过域随机化 (DR) 和课程学习来强化策略。残余差异在部署期间通过在线自适应进行管理。失配的典型来源包括驱动与机器人动力学、感知与状态估计，以及接触与地形建模。最终目标是实现鲁棒、自适应且可扩展的双足运动。

将执行器理想化为完美的力矩源，忽略了对于运动至关重要的现实世界非线性因素。实际上，实际传递的力矩取决于若干非理想效应：(i) 硬件限制，包括由有限电压、电流和热边界 [18] 引起的饱和与压摆率约束；以及 (ii) 机电动力学，包括有效过滤输入信号的有限带宽和指令延迟。

这些效应会同时改变接触力的时序和大小，这也是在仿真中看似稳定的策略在硬件上常常失败的主要原因。底层电机动力学由公式2描述：

$$\dot{i} = \frac{1}{L}(v - Ri - k_e \omega_m), \tau_m = k_t i, P = v i, \quad (2)$$

其中，施加的电压 v 和反电动势(EMF) $k_e \omega_m$ 决定了电流 i 以及由此产生的电机扭矩 τ_m 。如公式2所示，这些动力学对扭矩指令起到低通滤波器的作用，引入了延迟并限制了峰值响应能力。在仿真中忽略这种滤波效应，往往会鼓励策略变得过于激进或时序不当，从而在部署到物理机器人上时变得不稳定。因此，准确的迁移需要对执行器限制和根据经验测量校准的响应动力学进行显式建模。

b) 接触与地形建模：除了机器人动力学之外，仿真到现实差距的另一个主要来源是环境交互。对于双足运动而言，策略稳定性对建模的接触物理高度敏感。即使是接触刚度、摩擦或地形几何中的微小误差，也可能改变地面反作用力和步态时序，从而破坏原本有效的策略稳定性 [19][20]。此外，现实世界中的表面会因磨损、湿气或表面纹理而表现出摩擦和柔顺性的非平稳变化——这些因素大多被静态仿真模型所忽略。

变化——这些因素大多被静态仿真模型所忽略。

仿真器通常采用两种范式之一：计算成本高昂的刚体模型，能够捕捉尖锐的粘滑过渡；或者更稳定但精度较低的柔顺模型，通过弹簧-阻尼系统近似接触。两种方法都会引入伪影。调校不当的柔顺模型可能夸大冲击力或人为地阻尼能量，而简化的摩擦模型则可能使对平衡至关重要的压力中心 (CoP) 估计产生偏差。由于逼真的可变形地形仿真计算成本高昂，训练通常局限于刚性平面表面，导致策略在真实世界的柔顺地形（如地毯或草地）上失效。

c) 感知与状态估计：在仿真中，运动策略通常使用特权、无噪声的状态信息进行训练，包括真实的关节角度、速度和基座位姿。然而，在硬件上，这些量只能通过编码器、惯性测量单元和接触传感器，经过滤波和传感器融合流水线获得的估计值来获取。此类流水线会引入量化、偏置漂移、有色噪声、延迟、时间戳抖动以及偶发的丢包，使系统实际上变为部分可观测。由此产生的估计误差和延迟会扰动步态相位时序和反馈增益，导致在完美状态信息上训练的策略在部署时变得脆弱。

为缓解这些差异，训练设置应尽可能贴近真实观测接口与时序。类估计器输入应替代真实状态，感知与滤波的统计特性（噪声、偏差、延迟、丢包）需在物理合理范围内进行模拟。策略架构应整合信息

over time, for instance through short observation histories or recurrence, while any privileged variables are reserved for training-only supervision (e.g., critics or auxiliary losses) and excluded from the deployed actor. Maintaining end-to-end consistency between simulated and real observation pipelines—including units, timestamps, normalization, and filter warm-up behavior—substantially improves robustness to sensing imperfection [15].

d) Numerical Solvers: The choice of numerical solver presents a critical trade-off between simulation fidelity, numerical stability, and computational efficiency [21]. Solver approximations directly contribute to the sim-to-real gap by introducing numerical artifacts that can distort or amplify existing modeling errors. These effects manifest primarily in robot dynamics, where integrator type and step size influence stability and energy conservation, and in contact resolution, where the solver determines the realism of impacts and frictional interactions.

Solver design also shapes several system-level properties that are important for reinforcement learning, including determinism across runs, gradient quality during optimization, and scalability on parallel hardware [22]. While more accurate solvers improve physical fidelity, they also require greater computational resources, necessitating a careful balance between accuracy and throughput for modern DRL frameworks [21].

C. Pre-training Alignment

Simply increasing simulator fidelity is rarely sufficient to close the sim-to-real gap, since inevitable parameter uncertainties and unmodeled dynamics remain. To address these residual discrepancies, **pre-training alignment** is introduced as a suite of strategies applied before policy training to systematically improve model accuracy and reduce the burden on the learning algorithm.

a) Offline System Identification: System Identification (SI) enhances simulator fidelity by calibrating its physical parameters before policy training, thereby reducing the need for the learned policy to compensate for modeling errors. The procedure is typically formulated as a risk minimization problem. Using data collected from real-world experiments—often designed as “excitation trajectories” to expose model inaccuracies—an optimization algorithm searches for the parameters θ (and, optionally, a system delay Δ) that best reproduce the observed robot behavior. The objective minimizes a loss function that quantifies the discrepancy between simulated and real outputs, such as joint kinematics and contact forces.

$$\begin{aligned} \theta^*, \Delta^* &= \arg \min_{\theta \in \Theta, \Delta} J(\theta, \Delta) \\ &= \sum_{k=1}^{N_c} \sum_{t=0}^{H_k-1} \ell(y_{t,k}^{\text{real}}, \hat{y}_{t,k}^{\text{sim}}(\theta, \Delta)) + \|\theta - \theta_0\|_{W_\theta}^2. \end{aligned} \quad (3)$$

The scope of SI spans inertial, actuation, contact, and sensing parameters, as summarized in Table I. A successful SI workflow produces two primary outcomes: a calibrated nominal parameter set $\hat{\theta}$ for a higher-fidelity simulator, and

uncertainty estimates that provide principled bounds for DR during policy training.

TABLE I
SYSTEM IDENTIFICATION PARAMETERS IN BIPEDAL ROBOT SIM-TO-REAL TRANSFER.

Category	Parameters
Inertial	Mass; CoM; inertia tensor; link geometry; payload
Actuation	Torque constant; gear ratio; friction; backlash; compliance; delay; saturation
Contact	Stiffness; damping; friction coefficient; foot sole compliance; toe/heel joints; terrain properties
Sensing	Noise; bias; drift; latency; resolution; extrinsic calibration; FT calibration

In practice, the SI process often combines analytical measurements with whole-body TO. The workflow typically begins with inertial calibration to correct deviations in the CAD model, a procedure that may be conducted both before and after initial policy training to incorporate task-relevant data [23]. More critically, hardware-specific properties must be identified. For instance, reproducing realistic ground contact for the Cassie biped required experiments to determine the unique stiffness and damping characteristics of its compliant feet [24]. Similarly, characterizing actuation dynamics—such as motor constants and friction—is essential for matching the robot’s torque response and achieving stable gaits [19].

b) Residual Dynamics Learning: Residual discrepancies caused by unmodeled effects such as nonlinear friction, contact dynamics, or actuator thermal variations inevitably remain after initial system identification. Residual Dynamics Learning provides a complementary, model-centric approach to address this remaining gap. Instead of re-identifying all of the simulator’s base parameters, this method focuses on learning a direct correction to the simulator’s dynamics model to compensate for residual mismatches:

$$f_{\text{real}}(s_t, a_t) \approx f_{\text{sim}}(s_t, a_t) + g_\phi(s_t, a_t). \quad (4)$$

where s_t and a_t denote the state and action at time t . The function $g_\phi(s_t, a_t)$, typically parameterized by a neural network with parameters ϕ , outputs a predicted correction Δs_t to the simulator’s next-state prediction.

For example, [25] proposed a Residual Physics Learning framework that augments traditional system identification with a learned residual dynamics term. In their approach, the nominal simulator captures coarse rigid-body and buoyancy effects, while a neural residual model explains unmodeled forces and contacts. In contrast, a policy-centric formulation learns a residual directly in the action space. Instead of correcting the dynamics model, methods such as ASAP learn a delta-action model that directly adjusts the output of a simulation-trained policy [26].

D. Training for Robustness

While pre-training alignment seeks to reduce the sim-to-real gap by improving the simulator or performing initial policy corrections, an alternative and complementary philosophy is to train a policy that is inherently robust to this gap

随时间变化, 例如通过短时观测历史或递归实现, 同时所有特权变量仅保留用于仅训练监督 (如评论家或辅助损失), 并排除在部署执行器之外。保持模拟与真实观测管道之间的端到端一致性——包括单位、时间戳、归一化及滤波器预热行为——能显著提升对感知不完美 [15]的鲁棒性。

d) 数值求解器: 数值求解器的选择在仿真保真度、数值稳定性和计算效率之间呈现关键权衡 [21]。求解器近似会直接导致仿真到现实差距, 因为它引入的数值伪影可能扭曲或放大现有的建模误差。这些影响主要体现在机器人动力学中 (积分器类型和步长影响稳定性与能量守恒) 以及接触分辨率中 (求解器决定冲击和摩擦交互的真实感)。

求解器设计还塑造了多个对强化学习至关重要的系统级特性, 包括运行间的确定性、优化过程中的梯度质量以及并行硬件可扩展性 [22]。虽然更精确的求解器能提升物理保真度, 但它们也需要更多的计算资源, 因此需要在精度与吞吐量之间进行谨慎平衡, 以适应现代深度强化学习框架 [21]。

C. 预训练对齐

单纯提高仿真器保真度通常不足以弥合仿真到现实差距, 因为不可避免的参数不确定性和未建模动力学依然存在。为应对这些残余差异, **预训练对齐**被引入作为一套在策略训练前应用的策略, 旨在系统性地提升模型精度并减轻学习算法的负担。

a) 离线系统辨识: 系统辨识通过在策略训练前校准仿真器的物理参数来提升仿真器保真度, 从而减少学习策略对建模误差的补偿需求。该过程通常被表述为一个风险最小化问题。利用从真实世界实验 (通常设计为 “激励轨迹” 以暴露模型不准确性) 中收集的数据, 优化算法会搜索参数 θ (以及可选的系统延迟 Δ), 以最佳地复现观测到的机器人行为。目标是最小化一个损失函数, 该函数量化了仿真输出与真实输出 (如关节运动学和接触力) 之间的差异。

$$\begin{aligned} \theta^*, \Delta^* &= \arg \min_{\theta \in \Theta, \Delta} J(\theta, \Delta) \\ &= \sum_{k=1}^{N_c} \sum_{t=0}^{H_k-1} \ell(y_{t,k}^{\text{real}}, \hat{y}_{t,k}^{\text{sim}}(\theta, \Delta)) + \|\theta - \theta_0\|_{W_\theta}^2. \end{aligned} \quad (3)$$

系统辨识的范围涵盖惯性、驱动、接触和感知参数, 如表I所示。一个成功的系统辨识工作流会产生两个主要成果: 为高保真仿真器校准的标称参数集 θ , 以及

为策略训练中的域随机化提供原则性界限的不确定性估计。

表I 双足机器人仿真到现实迁移中的系统辨识参数。

类别	参数
惯性	质量; 质心; 惯性张量; 连杆几何; 负载
驱动	转矩常数; 齿轮比; 摩擦; 间隙; 柔顺性; 延迟; 饱和
接触	刚度; 阻尼; 摩擦系数; 脚底柔顺性; 脚趾/脚跟关节; 地形属性
感知	噪声; 偏差; 漂移; 延迟; 分辨率; 外参标定; 力/力矩传感器标定

在实践中, 系统辨识过程通常将分析测量与全身轨迹优化相结合。该工作流程通常从惯性标定开始, 以修正 CAD模型中的偏差, 该过程可在初始策略训练前后进行, 以纳入任务相关数据 [23]。更为关键的是, 必须辨识硬件特定属性。例如, 为Cassie双足机器人再现真实的地面接触, 需要通过实验确定其柔顺脚独特的刚度和阻尼特性 [24]。类似地, 表征驱动动力学 (如电机常数和摩擦) 对于匹配机器人的转矩响应并实现稳定步态至关重要 [19]。

b) 残差动力学学习: 在初始系统辨识之后, 由非线性摩擦、接触动力学或执行器热变化等未建模效应引起的残余差异不可避免地仍然存在。残差动力学学习提供了一种互补的、以模型为中心的方法来解决这一剩余差距。该方法并非重新辨识模拟器的所有基础参数, 而是专注于学习对模拟器动力学模型的直接修正, 以补偿残余失配:

$$f_{\text{real}}(s_t, a_t) \approx f_{\text{sim}}(s_t, a_t) + g_\phi(s_t, a_t). \quad (4)$$

其中 s_t 和 a_t 分别表示时间 t 时的状态和动作。函数 $g_\phi(s_t, a_t)$ 通常由参数为 ϕ 的神经网络参数化, 输出对仿真器下一状态预测的修正量 $\Delta s_{t\circ}$ 。

例如, [25] 提出了一种残差物理学习框架, 该框架通过一个学习型残差动力学项来增强传统的系统辨识。在他们的方法中, 名义仿真器捕捉粗略的刚体动力学和浮力效应, 而神经残差模型则解释未建模的力和接触。相比之下, 以策略为中心的公式直接在动作空间中学习残差。与修正动力学模型不同, ASAP 等方法学习一个增量动作模型, 该模型直接调整仿真训练策略 [26]的输出。

D. 鲁棒性训练

预训练对齐旨在通过改进仿真器或执行初始策略修正来缩小仿真到现实差距, 而另一种互补的理念则是从一开始就训练一个对该差距具有内在鲁棒性的策略。

from the outset. The primary objective of these in-training strategies is to enable zero-shot sim-to-real transfer, where the learned policy can be deployed directly to physical hardware without any real-world fine-tuning. This section discusses two major paradigms that address these challenges: DR and teacher–student learning.

a) Domain Randomization: To account for the inevitable discrepancies between simulation and reality, DR trains a policy across a wide distribution of simulated environments rather than a single deterministic one. The core idea is that if the randomized distribution is sufficiently broad to include the real world’s dynamics as one of its instances, the learned policy should transfer successfully without fine-tuning. This approach promotes robustness to variations but can also yield more conservative behaviors and requires significantly more training data.

In practice, DR involves randomizing a set of parameters at the start of each training episode. These parameters typically fall into two categories:

- 1) Dynamics parameters: Physical properties such as link masses and inertias, joint friction, motor strength, and actuator delays are varied within predefined ranges.
- 2) Environmental conditions: Factors such as random external forces, variations in initial states, sensor noise, latency, and ground friction are randomized to simulate real-world uncertainty.

The choice of randomization ranges is critical. Ranges that are too narrow may fail to cover the sim-to-real gap, while overly broad ranges can make the control problem intractable and lead to overly conservative or unstable policies.

DR has proven highly effective for sim-to-real transfer in legged robots. For example, controllers for the ANYmal quadruped have successfully employed a loose calibration plus DR strategy, in which core parameters are identified while difficult-to-model properties such as friction are heavily randomized [27]. Similarly, the DeepWalk policy learned to stabilize the NimbRo-OP2X biped by training with random external pushes, allowing the real robot to withstand disturbances [28].

To mitigate the training difficulty associated with wide randomization, Curriculum Learning (CL) is often used as a complementary strategy. Rather than exposing the policy to the full range of variations from the start, the agent is trained on progressively harder tasks. This can involve gradually widening the randomization ranges or increasing the target velocity, as demonstrated in the ALLSTEPS framework for the Cassie biped [6].

b) Teacher-student Paradigm: Beyond physical uncertainty, another central challenge in sim-to-real transfer is perceptual uncertainty. An agent in simulation has access to perfect, global, privileged information, such as the precise CoM velocity, ground friction coefficient, or exact contact forces. A real robot, however, must rely on noisy, delayed, and incomplete onboard sensors for state estimation. Training a policy directly across this large information gap often leads to failure.

The teacher–student paradigm is a knowledge distillation technique specifically designed to address this issue. The core

idea is to train two policies in simulation: (i) the teacher policy, π_{teacher} , which has access to all privileged information and can therefore learn near-optimal, expert-level behavior; and (ii) the student policy, π_{student} , which is also trained in simulation but is limited to the non-privileged sensory inputs that a real robot would have, such as simulated IMU readings, joint encoder angles, and camera images.

During training, the student does not learn directly from environmental interaction. Instead, it learns to reproduce the teacher’s actions through supervised learning, minimizing the discrepancy between its predicted actions and those generated by the teacher. The student thus learns a mapping function that uses only partial and noisy observations to approximate the expert’s decision-making process.

Ultimately, the student policy is deployed on the real robot, as it has been trained to perform expert-level behaviors under realistic perceptual constraints.

E. Post-deployment Adaptation

Even with high-fidelity simulation, offline system identification, and DR, learned policies are typically robust only within the distribution of conditions encountered during training. They perform well in environments that closely resemble the training domain, but real-world deployment inevitably introduces gradual drifts such as changes in ground friction, actuator heating, payload variations, or mechanical wear. These factors create residual mismatches and can degrade performance over time.

Online adaptation addresses this issue by adjusting policy behavior during execution using real-time feedback from robot–environment interactions. Approaches in this category can generally be divided into two main families: system identification–based adaptation, which reduces residual mismatches by either explicitly estimating time-varying physical parameters (for example, inertia, actuation, contact, and sensing) or implicitly inferring latent context variables from short observation–action histories; and adaptive policy learning, which directly adjusts the control policy through online fine-tuning, residual reinforcement learning, or other feedback-driven mechanisms.

a) Explicit Online System Identification: Online system identification can be broadly divided into two families: explicit approaches, which estimate physically meaningful parameters, and implicit approaches, which infer latent context representations from short interaction histories. This section introduces both perspectives and outlines their mechanisms, mathematical formulations, and representative applications in bipedal locomotion.

Explicit approaches directly estimate time-varying physical parameters such as inertia, CoM, actuation constants, contact stiffness, and sensor biases, allowing the underlying model or controller to remain calibrated during deployment. Common techniques include recursive least squares (RLS), disturbance observers, and extended or unscented Kalman filters (EKF/UKF). Formally, the estimation process can be expressed as

$$\hat{\phi}_t = E_{\psi}(h_t), \quad a_t \sim \pi_{\theta}(s_t; \hat{\phi}_t),$$

这些训练中策略的主要目标是实现零样本仿真到现实迁移, 使学习策略能够直接部署到物理硬件上, 无需任何现实世界微调。本节讨论应对这些挑战的两大范式: 域随机化与教师-学生学习。

a) 域随机化: 为了应对仿真与现实之间不可避免的差异, 域随机化 (DR) 并非在单一确定性环境中训练策略, 而是在广泛分布的仿真环境中进行训练。其核心思想是, 如果随机化的分布足够广泛, 能够将现实世界的动力学作为其一个实例包含在内, 那么学习型策略无需微调即可成功迁移。这种方法能提升对变化的鲁棒性, 但也可能导致更保守的行为, 并且需要显著更多的训练数据。

在实践中, 域随机化在每个训练回合开始时随机化一组参数。这些参数通常分为两类:

- 1) 动力学参数: 诸如连杆质量和惯性、关节摩擦、电机强度以及执行器延迟等物理属性在预定义范围内变化。
- 2) 环境条件: 诸如随机外力、初始状态变化、传感器噪声、延迟和地面摩擦等因素被随机化, 以模拟现实世界的不确定性。

随机化范围的选择至关重要。范围过窄可能无法覆盖仿真到现实差距, 而范围过宽则可能使控制问题变得棘手, 并导致策略过于保守或不稳定。

域随机化已被证明在足式机器人的仿真到现实迁移中非常有效。例如, ANYmal四足机器人的控制器成功采用了松散标定加域随机化策略, 即核心参数被识别, 而摩擦等难以建模的属性则被大幅随机化 [27]。类似地, DeepWalk策略通过训练时施加随机外部推力学会了稳定 NimbRo-OP2X双足机器人, 使真实机器人能够承受干扰 [28]。

为缓解宽范围随机化带来的训练难度, 课程学习常被用作补充策略。智能体并非从一开始就暴露于全部变化范围, 而是逐步接受难度递增的任务训练。这可以包括逐渐扩大随机化范围或提高目标速度, 正如ALLSTEPS框架在Cassie双足机器人上所展示的那样 [6]。

b) 师生范式: 除了物理不确定性, 仿真到现实迁移中的另一个核心挑战是感知不确定性。仿真中的智能体可以获取完美、全局的特权信息, 例如精确的质心速度、地面摩擦系数或准确的接触力。然而, 真实机器人必须依赖嘈杂、延迟且不完整的机载传感器进行状态估计。直接跨越这一巨大信息差距训练策略往往会导致失败。

师生范式是一种专门为解决此问题而设计的知识蒸馏技术。其核心

思想是在仿真中训练两个策略: (i) 教师策略 π_{teacher} , 它可以访问所有特权信息, 因此能够学习近乎最优的专家级行为; (ii) 学生策略 π_{student} , 它也在仿真中训练, 但仅限于真实机器人会拥有的非特权感知输入, 例如模拟的惯性测量单元读数、关节编码器角度和相机图像。

在训练过程中, 学生并不直接从环境交互中学习。相反, 它通过监督学习来复现教师的动作, 最小化其预测动作与教师生成动作之间的差异。因此, 学生学会了一个映射函数, 该函数仅利用部分且有噪声的观测来近似专家的决策过程。

最终, 学生策略被部署到真实机器人上, 因为它已经过训练, 能够在现实的感知约束下执行专家级行为。

E. 部署后自适应

即使采用高保真仿真、离线系统辨识和域随机化, 学习策略通常也仅对训练中遇到的条件分布具有鲁棒性。它们在高度相似于训练域的环境中表现良好, 但真实世界部署不可避免地会引入逐渐漂移, 例如地面摩擦变化、执行器发热、负载变化或机械磨损。这些因素会产生残余失配, 并可能随时间推移导致性能下降。

在线自适应通过在执行过程中利用机器人-环境交互的实时反馈来调整策略行为, 从而解决这一问题。该类方法通常可分为两大分支: 基于系统辨识的自适应, 通过显式估计时变物理参数 (例如惯性、驱动、接触和感知) 或从短时观测-动作历史中隐式推断潜在上下文变量来减少残余失配; 以及自适应策略学习, 通过在线微调、残余强化学习或其他反馈驱动机制直接调整控制策略。

a) 显式在线系统辨识: 在线系统辨识可大致分为两类: 显式方法 (估计具有物理意义的参数) 和隐式方法 (从短期交互历史中推断潜在上下文表征)。本节将介绍这两种视角, 并概述其机制、数学公式以及在双足运动中的代表性应用。

显式方法直接估计时变物理参数, 如惯性、质心、驱动常数、接触刚度和传感器偏差, 使得底层模型或控制器在部署过程中保持校准。常用技术包括递归最小二乘 (RLS)、扰动观测器以及扩展或无迹卡尔曼滤波器 (EKF/UKF)。形式上, 估计过程可以表示为

$$\hat{\phi}_t = E_{\psi}(h_t), \quad a_t \sim \pi_{\theta}(s_t; \hat{\phi}_t),$$

where $\hat{\phi}_t$ denotes the online estimate of the physical parameters, h_t is a short trajectory segment, and π_θ is the policy conditioned on the updated parameters.

Representative examples include Kalman filter–based estimation of inertial and contact properties in bipedal locomotion [29], as well as recent work on physically consistent EKF adaptation for humanoid loco-manipulation [30].

b) Implicit Online System Identification: Implicit approaches do not attempt to recover the physical parameters ϕ explicitly. Instead, they infer a compact latent variable from recent observation–action histories and condition the policy on it to achieve online adaptation. Formally,

$$z_t = g_\psi(h_t), \quad a_t \sim \pi_\theta(o_t, z_t), \quad h_t = (o_{t-H:t}, a_{t-H:t-1}).$$

Compared with explicit online system identification, which estimates ϕ and updates controller gains, the implicit approach amortizes system identification into the encoder g_ψ and emphasizes rapid, task-relevant adaptation.

A useful taxonomy distinguishes three common ways to obtain the context variable z_t from observation and action histories. (i) Recurrent state update: maintain an online latent state $z_t = f_\psi(z_{t-1}, o_t, a_{t-1})$, where f_ψ is implemented using a gated recurrent unit, long short-term memory network, or a learned state-space update. (ii) Sliding-window encoding: construct $h_t = (o_{t-H:t}, a_{t-H:t-1})$ and map it through a temporal encoder $z_t = g_\psi(h_t)$; common choices include temporal convolutions or causal attention encoders. (iii) Latent filtering: treat z_t as a belief state with learned prior and posterior distributions, $p_\psi(z_t | z_{t-1}, a_{t-1})$ and $q_\psi(z_t | z_{t-1}, a_{t-1}, o_t)$, updated in a Kalman-like manner and trained using predictive or variational objectives. In practice, z_t is often smoothed using exponential moving averages and temporally regularized; resets are triggered when prediction residuals or $\|\Delta z_t\|$ exceed predefined thresholds.

Memory-based approaches with recurrent policies provide strong evidence that encoding observation and action histories enables adaptation under partial observability. For example, an end-to-end LSTM policy was transferred to the biped Cassie in zero-shot fashion [2]. Rapid Motor Adaptation (RMA) combines a base locomotion policy with an adaptation module that infers a low-dimensional context from recent histories, allowing quadruped robots to adapt in real time across diverse terrains [31]. Building on this idea, A-RMA extends the method to bipedal locomotion, where the context-conditioned policy enables fast adaptation to changing loads and terrains [32]. More recently, transformer-based in-context adaptation has scaled such history-conditioned mechanisms to humanoid robots [15].

IV. DISCUSSION

The sim-to-real approaches reviewed in this chapter are not isolated solutions but interconnected components of a unified strategy, organized by their application stage from pre-training alignment to post-deployment adaptation. The success of this strategy rests on two complementary philosophies: reducing the sim-to-real gap through higher model fidelity and making the policy resilient to the errors that inevitably remain. This

discussion synthesizes these ideas, outlining how these tools work together in practice, where each method helps, where it falls short, and what is needed to advance the field.

A. Sim-to-Real Methods Limits

While each sim-to-real technique provides a powerful solution to a specific aspect of the sim-to-real gap, a critical analysis reveals that no single method serves as a universal remedy. In practice, relying exclusively on one approach exposes a distinct set of limitations. A clear understanding of these individual shortcomings is therefore essential for navigating the complex landscape of sim-to-real transfer.

Challenges already arise at the pre-training alignment stage. System identification is essential but labor-intensive, and it struggles to capture complex, unmodeled dynamics or slow nonstationary drift due to wear or recalibration. Residual learning is often introduced to compensate, yet it introduces a different trade-off: dependence on real-world data collection, which is costly and safety-constrained, and corrections that are narrow in scope, often failing to generalize beyond the training envelope or to new tasks.

Similarly, training-for-robustness strategies bring their own compromises. DR, the cornerstone of zero-shot transfer, can produce overly conservative policies that sacrifice peak performance for generality. Moreover, selecting effective randomization ranges remains a poorly scaled, heuristic-driven process. The teacher–student paradigm, while effective for addressing perceptual uncertainty, is limited by the teacher’s maximum achievable performance and does not, by itself, resolve mismatches in physical dynamics.

Post-deployment adaptation can handle long-term changes but introduces the classic stability–plasticity dilemma: the agent must adapt without catastrophically forgetting prior knowledge. This adds complexity and raises safety concerns, since a continually adapting policy is inherently non-stationary and difficult to verify. In short, every paradigm imposes trade-offs—whether in data dependency, performance conservatism, or algorithmic complexity. The most practical solutions arise not from selecting a single method but from intelligently integrating multiple approaches.

B. From Theory to Practice

The most effective sim-to-real solutions emerge from the synergistic combination of complementary methods rather than reliance on any single technique. A dominant strategy is to pair a high-fidelity baseline with robust training. This typically involves first applying offline system identification or residual dynamics learning to construct a more accurate simulator, which then serves as the foundation for training more robust policies [25]. This initial alignment allows for narrower and more targeted randomization, mitigating the policy conservatism often associated with DR.

A second, complementary paradigm produces policies that are robust to both perceptual limitations and dynamic environmental variations. This is achieved by combining the teacher–student framework, which hardens the policy against sensor noise, with implicit online system identification [4],

其中 ϕ_t 表示物理参数的在线估计值, h_t 是一个短轨迹段, π_θ 是基于更新参数的条件策略。

代表性示例包括基于卡尔曼滤波器的双足运动惯性与接触属性估计 [29], 以及近期关于人形机器人移动操作中物理一致性扩展卡尔曼滤波器自适应的工作 [30].

b) 隐式在线系统辨识: 隐式方法不试图显式恢复物理参数 ϕ 。相反, 它们从最近的观测-动作历史中推断出一个紧凑的潜变量, 并以此条件化策略来实现在线自适应。形式上,

$$z_t = g_\psi(h_t), \quad a_t \sim \pi_\theta(o_t, z_t), \quad h_t = (o_{t-H:t}, a_{t-H:t-1}).$$

与估计 ϕ 并更新控制器增益的显式在线系统辨识相比, 隐式方法将系统辨识分摊到编码器 g_ψ 中, 并强调快速、任务相关的适应。

一个有用的分类法区分了从观测和动作历史中获取上下文变量 z_t 的三种常见方式。 (i) 循环状态更新: 维护一个在线潜状态 $z_t = f_\psi(z_{t-1}, o_t, a_{t-1})$, 其中 f_ψ 使用门控循环单元、长短期记忆网络或学习型状态空间更新实现。 (ii) 滑动窗口编码: 构建 $h_t = (o_{t-H:t}, a_{t-H:t-1})$ 并通过时间编码器 $z_t = g_\psi(h_t)$ 映射; 常见选择包括时间卷积或因果注意力编码器。 (iii) 潜滤波: 将 z_t 视为具有学习型先验和后验分布的信念状态, 即 $p_\psi(z_t | z_{t-1}, a_{t-1})$ 和 $q_\psi(z_t | z_{t-1}, a_{t-1}, o_t)$, 以类卡尔曼方式更新, 并使用预测或变分目标进行训练。 在实践中, z_t 通常使用指数移动平均进行平滑并施加时间正则化; 当预测残差或 $\|\Delta z_t\|$ 超过预定义阈值时触发重置。

基于记忆的方法结合循环策略提供了有力证据, 表明编码观测和动作历史能够在部分可观测性下实现适应。例如, 一个端到端LSTM策略以零样本方式迁移到了双足机器人Cassie上 [2]。快速运动自适应 (RMA) 将基础运动策略与一个自适应模块相结合, 该模块从近期历史中推断出低维上下文, 使四足机器人能够在多样化地形上实时适应 [31]。在此基础上, A-RMA将该方法扩展到双足运动, 其中基于上下文的策略能够快速适应变化负载和地形 [32]。最近, 基于Transformer的上下文自适应将这种基于历史条件的机制扩展到了人形机器人 [15]。

IV. 讨论

本章所综述的仿真到现实方法并非孤立的解决方案, 而是统一策略中相互关联的组成部分, 按其应用阶段从预训练对齐到部署后自适应进行组织。该策略的成功依赖于两种互补的理念: 通过更高的模型保真度缩小仿真到现实差距, 以及使策略能够对不可避免残留的误差具有鲁棒性。这一

讨论综合了这些观点, 概述了这些工具如何在实践中协同工作, 每种方法的优势所在、其不足之处, 以及推动该领域发展所需的条件。

A. 仿真到现实方法的局限性

尽管每种仿真到现实技术都为仿真到现实差距的特定方面提供了强大的解决方案, 但批判性分析表明, 没有一种单一方法能作为通用解决方案。在实践中, 仅依赖单一方法会暴露出一系列独特的局限性。因此, 清晰理解这些各自的不足, 对于应对仿真到现实迁移的复杂局面至关重要。

挑战在预训练对齐阶段就已显现。系统辨识至关重要但劳动密集, 且难以捕捉复杂的未建模动力学或因磨损或重新校准导致的缓慢非平稳漂移。残差学习常被引入以进行补偿, 但这又带来了不同的权衡: 依赖真实世界数据采集 (成本高昂且受安全约束), 以及修正范围狭窄, 往往无法泛化到训练范围之外或新任务。

同样, 鲁棒性训练策略也带来了自身的妥协。作为零样本迁移基石的域随机化, 可能产生过于保守的策略, 为追求通用性而牺牲峰值性能。此外, 选择有效的随机化范围仍然是一个难以规模化、依赖启发式方法的过程。师生范式虽能有效应对感知不确定性, 但受限于教师策略的最大可达性能, 且其本身无法解决物理动力学失配问题。

部署后自适应可以处理长期变化, 但引入了经典的稳定性-可塑性困境: 智能体必须在适应过程中避免灾难性地遗忘先前知识。这增加了复杂性并引发了安全担忧, 因为持续自适应的策略本质上具有非平稳性且难以验证。简而言之, 每种范式都带来了权衡——无论是数据依赖性、性能保守性还是算法复杂性。最实用的解决方案并非来自选择单一方法, 而是来自智能地整合多种方法。

B. 从理论到实践

最有效的仿真到现实解决方案源于互补方法的协同组合, 而非依赖单一技术。一种主流策略是将高保真基线模型与鲁棒训练相结合。这通常首先应用离线系统辨识或残差动力学学习来构建更精确的仿真器, 进而作为训练更鲁棒策略的基础 [25]。这种初始对齐使得域随机化范围更窄、更具针对性, 从而缓解了域随机化常伴随的策略保守性问题。

第二种互补范式能够生成对感知限制和动态环境变化均具有鲁棒性的策略。这是通过将师生框架 (该框架使策略对传感器噪声具有鲁棒性) 与隐式在线系统辨识 [4], 相结合来实现的。

[15]. In this setup, an encoder uses the robot’s recent history to infer the current real-world dynamics, allowing the policy to adapt its behavior online. This combination of robust perception and continuous adaptation forms the foundation of several state-of-the-art systems [20].

C. Future Outlook

The trajectory of sim-to-real development can be viewed as an evolutionary progression through three stages. The field has advanced from early Stage 1, Simulation plus Fine-tuning, which was hindered by the inefficiency and risks of real-world learning, to the current dominant paradigm, Stage 2, Zero-Shot Transfer. Although techniques such as DR have enabled much of today’s agile locomotion, the resulting policies are often conservative and brittle when faced with out-of-distribution conditions.

The next frontier is Stage 3, Real-World Mastery, in which robots progress beyond transferring a fixed skill set to continuously adapting and improving in open-world environments. This stage envisions agents that achieve genuine competence by integrating extensive simulated experience with rapid online learning, enabling them to handle novel situations long after their initial deployment.

The key challenge lies in transitioning from the static paradigm of Stage 2 to the dynamic paradigm of Stage 3. This shift requires moving beyond one-way transfer toward a continuous learning loop, supported by concurrent advances in three areas. It will require better models, evolving from static simulators into adaptive digital twins that are continuously updated with real-world data. It will require better policies, advancing from task-specific optimization toward generalization through improved algorithms and architectures. Above all, it will demand better adaptation, requiring online learning algorithms that are not only fast and data-efficient but also provably safe, establishing the foundation for true lifelong robotic learning.

V. CONCLUSIONS

Sim-to-real for bipedal locomotion is not a bag of tricks but a principled approach to modeling, learning, and adaptation. Whether this is implemented within an end-to-end framework or a multi-layered hierarchical scheme, the core challenges of bridging the sim-to-real gap remain the same. The most effective way to navigate these challenges is to focus on two primary lines of effort, reinforced by a continuous adaptation loop. The first line of effort targets the Model: shrinking the sim-to-real gap by raising simulator fidelity through careful identification. The second targets the policy: hardening it against residual errors by training for variability. The adaptation loop ensures that once deployed, the robot can continuously read the world and adjust to keep its behavior robust and safe.

In practice, this means focusing on the model first by calibrating a verifiable nominal model. The focus then shifts to the policy, training with measurement-driven DR and mirroring the real observation pipeline. Finally, the adaptation loop is closed by equipping the controller with mechanisms

for online inference and adjustment. Each of these efforts carries inherent trade-offs: identification is laborious, broad randomization can yield conservative policies, and adaptation complicates verification. A disciplined blend is therefore more reliable than any single method.

The outlook for the field is one of increasing integration and capability across all three fronts. Progress will depend on advances in simulation, policy design, and adaptation. Future simulators will evolve from static tools into continuously updated digital twins or generative world models that learn directly from real-world data. Policies will become more general, drawing on large-scale foundation models to move beyond task-specific training toward versatile, goal-directed competence. Adaptation mechanisms will grow more powerful, enabling online learning that is faster, safer, and more data-efficient. The guiding principle remains clear: begin with physics you can verify, train for the variability you expect, and empower the robot to learn as it walks.

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[15]. 在此设置中，编码器利用机器人的近期历史来推断当前的现实世界动力学，从而使策略能够在线调整其行为。这种鲁棒感知与持续适应的结合构成了多个最先进系统的基础 [20]。

C. 未来展望

仿真到现实发展的轨迹可被视为一个历经三个阶段的演进过程。该领域已从早期的第一阶段“仿真加微调”（受限于现实世界学习的低效与风险）发展到当前主导范式——第二阶段“零样本迁移”。尽管域随机化等技术已促成当今许多敏捷运动，但由此产生的策略在面对分布外条件时往往趋于保守且脆弱。

下一个前沿是第三阶段“现实世界精通”，在此阶段，机器人将超越迁移固定技能集，转而持续适应并改进于开放世界环境。这一阶段设想智能体通过整合广泛的仿真经验与快速的在线学习，实现真正的能力提升，使其在初始部署后仍能长期应对新情况。

关键挑战在于从第二阶段的静态范式过渡到第三阶段的动态范式。这一转变需要超越单向迁移，迈向持续学习循环，并依托三个领域的同步进展。这需要更好的模型——从静态模拟器演进为持续用现实数据更新的自适应数字孪生；需要更好的策略——通过改进算法与架构，从任务特定优化迈向泛化；最重要的是，需要更好的适应——要求在线学习算法不仅快速且数据高效，还需具备可证明的安全性，为真正的终身机器人学习奠定基础。

V. 结论

双足运动的仿真到现实并非一套技巧集合，而是一种关于建模、学习和适应的原则性方法。无论这一过程是在端到端框架还是多层分层方案中实现，弥合仿真到现实差距的核心挑战始终如一。应对这些挑战最有效的方式是聚焦于两条主要工作线，并由一个持续的适应循环加以强化。第一条工作线针对模型：通过精细辨识提升仿真器保真度，从而缩小仿真到现实差距。第二条工作线针对策略：通过变异性训练，使其对残差误差具有鲁棒性。适应循环确保机器人一旦部署，能够持续感知世界并调整自身行为，以保持其鲁棒性和安全性。

在实践中，这意味着首先聚焦于模型，通过校准一个可验证的标称模型。随后重点转向策略，利用测量驱动领域的随机化进行训练，并镜像真实观测流水线。最后，通过为控制器配备在线推理与调整机制，闭环适应循环。

这些努力各自都带有固有的权衡：辨识工作繁重，广泛随机化可能产生保守策略，而适应则增加了验证的复杂性。因此，有纪律的融合比任何单一方法都更为可靠。

该领域的前景是在所有三个前沿方向上实现日益增强的整合与能力。进展将依赖于仿真、策略设计和适应方面的突破。未来的仿真器将从静态工具演变为持续更新的数字孪生或直接从现实世界数据中学习的生成式世界模型。策略将变得更加通用，借助大规模基础模型，从任务特定训练迈向多功能、目标导向的能力。适应机制将变得更加强大，实现更快、更安全且数据效率更高的在线学习。指导原则依然明确：从可验证的物理原理出发，针对预期的变异性进行训练，并赋予机器人在行走中学习的能力。

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